

Residual Radial Structure in SPARC Dark Matter Halos Reproduces on Independently Derived Rotation Curves

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ABSTRACT

Single smooth equilibrium halo profiles (NFW, Burkert, Einasto) leave statistically significant, radially organized residuals in $\sim 35\%$ of SPARC rotation curves: a minimum-order radial-domain decomposition, with model order selected by the Bayesian Information Criterion, requires a second cored radial domain in that fraction of galaxies. Such a detection could reflect the flexibility of the fitting basis or a localized systematic in one velocity-field extraction rather than a property of the galaxies. We report the most direct test of galaxy-intrinsic origin: whether the multi-domain selection reproduces when the rotation curve is re-derived from the original radio data cubes by independent software, holding the baryonic model fixed so that only the measured velocities change. It does. In four of five adjudicable multi-domain systems the selection recovers — decisively for NGC 2403 ($\Delta\text{BIC} = 121$, on a curve derived from different observations than the SPARC curve) and NGC 3198 ($\Delta\text{BIC} = 55$), on a warp-aware curve for NGC 2841, and at lower significance on a different-telescope extraction for NGC 3741 — with the crossover radius preserved. The one non-reproduction and one power-limited case are reported explicitly, and single-domain controls reproduce their single-domain selection. Because a rotation curve constrains only the enclosed-mass profile $M(<r)$, we make no claim about the three-dimensional geometry or physical origin of the structure; the result is confined to a feature in the enclosed-mass profile that is reproducible under independent velocity derivation. A companion analysis (Bibb 2026b) presents the full method-independence and robustness program that further characterizes it.

Keywords: Dark matter (353) — Galaxy rotation curves (619) — Galaxy dark matter halos (1880) — Astrostatistics (1882)

1. INTRODUCTION

Rotation curves of late-type galaxies are conventionally decomposed into baryonic contributions plus a single smooth equilibrium dark matter halo — cuspy (NFW; Navarro et al. 1996), cored (Burkert; Burkert 1995), or intermediate (Einasto; Einasto 1965). The adequacy of a single smooth profile is usually assessed only through global fit quality, and coherent residual features at particular radii have been discussed largely in connection with baryonic structure (“Renzo’s rule”; Sancisi 2004). Applying a minimum-order radial-domain decomposition to SPARC rotation curves (Lelli et al. 2016) — the dark-matter contribution approximated by the smallest number of cored radial mass components the data support — we find that a second cored radial domain is required in $\sim 35\%$ of galaxies (§3). A companion

analysis (Bibb 2026b) establishes that this selection is robust to the fitting basis — stable across NFW, Burkert, and free-shape Einasto backbones, recovered by a structurally unrelated Gaussian-shell decomposition at consistent radii, and not predicted by galaxy mass or other measured properties.

Basis-independence, however, does not by itself exclude the possibility that the structure originates in the specific rotation curves published by SPARC — for example in a localized systematic of one velocity-field extraction — rather than in the galaxies. This Letter reports the single most stringent test of that possibility, and the most direct observational test of galaxy-intrinsic origin: reproduction of the multi-domain selection on rotation curves re-derived from the original radio data cubes by independent software. A fit improvement on one published curve, however constrained the basis, does not predict the outcome of an independent velocity derivation. Such a derivation changes the observable

being fitted, not merely the parameterization: agreement requires the inferred structure to survive independent measurement, data-reduction choices, and fitting simultaneously. That this agreement holds is what distinguishes a property of the galaxy from a property of the parameterization. We state the atomic result — the structure survives independent measurement — and defer the full validation program to the companion paper.

2. DATA AND METHODS

2.1. Sample, baryonic model, and decomposition

We use the SPARC database (Lelli et al. 2016), restricted to galaxies with quality flag $Q \leq 2$ and at least six rotation-curve points after removing points interior to the first reliably measured radius (121 converged fits). Baryonic contributions adopt fixed mass-to-light ratios $\Upsilon_{\text{disk}} = 0.5$ and $\Upsilon_{\text{bul}} = 0.7$ at $3.6 \mu\text{m}$ with the gas contribution at unity, combined in signed squares; velocity uncertainties are floored at 1 km s^{-1} . The inferred dark-matter contribution is approximated by a minimum-order sum of cored radial mass components of Burkert form, $V_{\text{DM}}^2(r) = \sum_{i=1}^N V_{\text{B}}^2(r; \rho_{0,i}, a_i)$, with $\rho_{0,i}$ strictly decreasing and a_i strictly increasing outward. Model order is selected by $\text{BIC} = \chi^2 + k \ln n_{\text{pts}}$ (Schwarz 1978; Liddle 2007), $k = 2N$, over $N = 1-4$; at $N = 1$ the model reduces exactly to a single Burkert halo. The capped-fit stability variant and the full basis-independence program are detailed in the companion paper (Bibb 2026b); the present Letter applies the same selection procedure and reports its outcome directly.

2.2. Independent velocity re-derivation

The reproduction test holds the baryonic model and the decomposition machinery fixed and replaces only the measured rotation curve, re-deriving it from the original radio data cubes by independent software. For NGC 2403 and NGC 3198 we re-derive curves directly from public THINGS-class cubes using independent tilted-ring forward-modeling (^{3D}Barolo; Di Teodoro & Fraternali 2015), with geometry fixed to literature tilted-ring solutions. For the strongly warped systems NGC 2841 and NGC 3521, a blind fixed-geometry extraction cannot test the feature; we instead use published warp-aware tilted-ring curves (DiskFit; Sellwood & Spekkens, private communication) from the same THINGS cubes. For NGC 3741 we use an independent VLA-ANGST extraction (a different telescope from the WSRT-based SPARC curve; Walter et al. 2008). Dwarf controls DDO 154 and DDO 168 use asymmetric-drift-corrected curves from an independent group (3D forward-modeling; Iorio et al. 2017). Each galaxy’s re-derived curve is passed through the identical selection

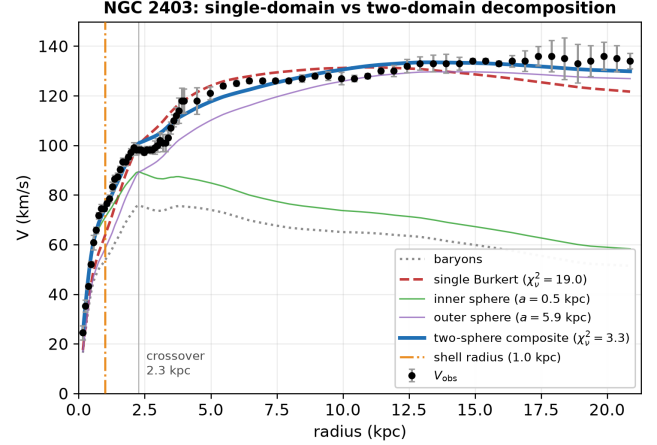


Figure 1. Representative two-domain decomposition (NGC 2403, SPARC). The observed rotation curve (points) and baryonic contribution (dotted) are shown with the best single-Burkert fit (dashed) and the BIC-selected two-domain composite (thick solid); the inner and outer cored components are shown individually (thin solid). The single profile cannot reproduce both the inner rise and the outer profile simultaneously, whereas the two-domain composite does (χ^2_{ν} improving from $\simeq 19$ to $\simeq 3$). NGC 2403 is the anchor of the present Letter: the same two-domain selection recovers at $\Delta\text{BIC} = 121$ on a rotation curve re-derived from the original radio cubes by independent software, and from different observations than those underlying the SPARC curve (§3).

procedure; a reproduction is a recovery of the same BIC-selected model order at a consistent crossover radius.

3. RESULTS

We find that a second radial domain is BIC-selected in $\sim 35\%$ of the 121 galaxies, defining the multi-domain population tested here; its full statistical characterization — robustness to the fitting basis, decision margins, and the systematic bias a single smooth fit imposes on the inferred core scale — is presented in the companion paper (Bibb 2026b). The present Letter asks the narrower question the reproduction test answers: when the measured velocities of these galaxies are independently re-derived, does the same population reappear? Figure 1 shows NGC 2403, the anchor case, in the SPARC data.

The central result of this Letter concerns whether that structure survives an independent velocity derivation. It does. NGC 2403 selects $N = 2$ at $\Delta\text{BIC} = 121$ with the domain crossover at 1.9 kpc (SPARC-based value $\simeq 2.3$ kpc), on a curve that also derives from different observations than those underlying the SPARC curve; NGC 3198 selects $N = 2$ at $\Delta\text{BIC} = 55$. The warp-aware curve for NGC 2841 selects $N = 2$ at $\Delta\text{BIC} = 11$ — its earlier failure on a blind extraction is attributable to the warp suppressing the feature, which the warp-

aware curve recovers. NGC 3741 reproduces at lower significance on the independent VLA-ANGST extraction, the modest signal-to-noise of this dwarf limiting the decision margin. The one non-reproduction, NGC 3521, prefers a single domain on its warp-aware curve ($\Delta\text{BIC} = -7$ against two) and is reported as such; an injection–recovery power control confirms this verdict is informative rather than power-limited (the feature it would need to show perturbs the curve well above the detectable amplitude). A single system, NGC 1003, is genuinely instrument-limited — its independent HALO-GAS extraction yields only 14 usable rings, and a power control shows the feature it would carry lies below the detectable amplitude — so its non-reproduction is uninformative. The two single-domain controls, DDO 154 and DDO 168, reproduce their single-domain selection.

The adjudicated tally is four reproductions (NGC 2403, NGC 3198, NGC 2841, NGC 3741) spanning three classes of independence (same cubes/different software, different observations, and different telescope), one honest non-reproduction (NGC 3521), one quantified power-limited case (NGC 1003), and two single-domain controls reproduced. Reproduction holds in four of the five adjudicable galaxies; the failure and the exclusion are reported explicitly rather than absorbed into an undifferentiated “instrument-limited” category. Because the re-derived velocities are an independent measurement of each galaxy — in the strongest cases from different observations or a different telescope — a recovery of the same model order at the same radius is evidence that the structure is consistent with being an intrinsic property of the galaxy rather than a feature of the particular published velocity extraction.

4. DISCUSSION

We frame this reproduction as proof of principle rather than population-level validation: the adjudicable sample is small, limited by the subset of SPARC galaxies whose rotation curves can be re-derived from public cubes at sufficient quality. It establishes that the multi-domain selection *can* reproduce under genuinely independent velocity derivation, not the rate at which it does across the full sample; the latter awaits wider cube availability. What it does establish is decisive for the interpretation of the phenomenon: the structure is not an artifact of the specific SPARC velocity extraction, because it survives when that extraction is replaced.

A boundary on interpretation is geometric and absolute. A rotation curve constrains only the enclosed-mass profile $M(<r)$ through $V^2(r) = GM(<r)/r$; it carries no information about how the mass producing a given $M(<r)$ is arranged in angle or in the third dimension.

A second radial domain, a spherical shell, a planar ring, a stream, a warp-induced redistribution, and a localized deposit are indistinguishable in $V(r)$ if they enclose the same mass within each radius. We therefore make no claim about the three-dimensional geometry of the detected structure, and none about its physical origin — whether baryonic (e.g. features propagated from the gas disk; Sancisi 2004), kinematic, or a property of the dark matter distribution. The present claim is confined to what rotation curves can support: a feature in the enclosed-mass profile reproducible under independent velocity derivation. Discriminating a centrally concentrated from a localized geometry requires observables that carry angular information — azimuthal structure in the velocity field or in resolved tracers — and is deferred to analyses of that kind.

The companion paper (Bibb 2026b) presents the full evidence that this structure is also *method*-independent: its stability across three halo families, its concordance with an unrelated Gaussian-shell decomposition at consistent radii (mass-adjusted odds ratio ~ 37), the forced-fit null that distinguishes selected structure from basis flexibility, the absorption of the localized shells by the broad two-domain model, and the negative results that bound the interpretation. The present Letter isolates the single element of that program that most directly answers “is it real”: the structure reproduces on an independent measurement of the same galaxies.

5. CONCLUSIONS

A minimum-order radial-domain decomposition selects a second radial domain in $\sim 35\%$ of SPARC rotation curves. Where the structure is strong enough to test, that selection reproduces on rotation curves re-derived from the original radio data cubes by independent software — in four of five adjudicable galaxies, decisively for NGC 2403 ($\Delta\text{BIC} = 121$) and NGC 3198 ($\Delta\text{BIC} = 55$), with the crossover radius preserved, and with the one non-reproduction and one power-limited case reported explicitly. The residual radial structure concealed by single smooth halo fits is therefore consistent with being an intrinsic property of the galaxies rather than a consequence of a particular published velocity extraction. As to its three-dimensional geometry and physical origin, the result is deliberately silent: a rotation curve constrains only the enclosed-mass profile, and those questions require observables that carry the angular information rotation curves lack.

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Photometry and Accurate Rotation Curves (SPARC) database (Lelli et al. 2016). This research was conducted independently, without institutional or grant support.

DATA AVAILABILITY

The decomposition code, the fitting and model-selection pipeline, the independent rotation-curve re-derivations, and the derived data products underlying this Letter are openly archived at Zenodo² and de-

veloped at <https://github.com/RonBibb/sparc-sphere>. The SPARC rotation curves are publicly available (Lelli et al. 2016); the independently re-derived curves are reproduced from public radio data cubes (THINGS, VLA-ANGST, HALOGAS) with the tilted-ring software cited in §2. All results are reproducible from the archived pipeline with the provided configuration.

Software: numpy, scipy, pandas, astropy, matplotlib, ^{3D}Barolo (Di Teodoro & Fraternali 2015)

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² <https://doi.org/10.5281/zenodo.21196421>